

No. of Functional Features Included in the Solution

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Voice Recording	Records user voice input for evaluation	Frontend (Streamlit)	Done	High
2	Audio Upload	Allows uploading speech audio files	Frontend	Done	High
3	Speech-to-Text	Converts speech into text using Whisper	Backend	Done	High
4	Semantic Similarity Analysis	Compares transcript with expected answer	Evaluation Module	Done	High
5	AI Feedback Generation	Generates personalized feedback using Gemini AI	AI Module	Done	High
6	Performance Dashboard	Displays scores and analytics	Dashboard	Done	Medium
7	PDF Report Generation	Creates downloadable evaluation reports	Report Module	Done	Medium
8	History Management	Stores previous evaluations in SQLite	Database	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Voice Recording, Audio Upload, Speech-to-Text, Semantic Similarity Analysis, AI Feedback Generation

Additional / Nice-to-Have Features	Dashboard Analytics, PDF Report Generation, Evaluation History
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Voice Recording, Audio Upload
2	Backend / Logic	3	Speech-to-Text, Similarity Analysis, AI Feedback
3	Database / Storage	1	SQLite Evaluation History
4	API / Integration	1	Google Gemini API Integration
5	Security / Authentication	0	Not Implemented